

Implemented Simulation Quality Report

A conservative, evidence-backed portfolio assessment of educational value, real-time visual quality, narration, interactivity, accessibility, stability and deployment readiness.

13

launchable simulations

75.7

portfolio average / 100

612

repository tests passing

85

packaged narration clips

AUDIT POSITION

All simulations remain Internal QA. Scores are product-readiness indicators based on repository and live-deployment evidence; they are not proof of classroom learning outcomes or signed Quest acceptance.

Strong learning architecture; uneven production readiness

The portfolio has meaningful educational interactions, but visual delivery, narration consistency and release verification are not yet uniform.

75.7

Average score

1

Pilot candidate

11

Promising Internal QA

1

Needs focused improvement

What management should know

PORTFOLIO STRENGTHS

- Educational design is the strongest layer: prediction, action, evidence and assessment are present across most lessons.
- The shared WebXR framework, comfort controls and 612-test repository provide a credible engineering base.
- Solubility is the clearest controlled-pilot candidate; pollination and states of matter are next in line after rendering acceptance.

RELEASE RISKS

- Solar System is currently blocked in production by a runtime error before the 3D canvas mounts.
- Digestive System requests 11 organ GLBs that all return 404, forcing a lower-fidelity procedural fallback.
- Only five simulations have packaged narration; eight depend on device-specific browser speech.
- No simulation has school outcome evidence or signed Quest acceptance, and Colour Adventure is missing from the canonical release registry.

RECOMMENDED PORTFOLIO DECISION

Do not present the complete portfolio as school-ready. Fix the two release blockers, verify the blank post-entry frames on real browsers and Quest, standardise narration, then pilot Solubility and Pollination in a controlled teacher-led setting.

A staged release protects trust: one or two validated flagship lessons are more valuable than thirteen uneven experiences promoted at the same maturity.

Ranked management scorecard

Scores combine eight weighted dimensions. A high educational score cannot fully offset a broken live route or missing production assets.

#	Simulation	Score	Band	Evidence
1	Soluble and Insoluble Substances Lab	92	Pilot candidate	High
2	Plant Pollination & Growth Cycle	82	Promising Internal QA	High
3	States of Matter Particle Lab	81	Promising Internal QA	Medium-high
4	Electric Circuits & Resistance (Ohm's Law)	79	Promising Internal QA	Medium-high
5	Introduction to Money	76	Promising Internal QA	Medium
6	Preposition Adventure	76	Promising Internal QA	Medium
7	Effects of Force on Motion and Shape	74	Promising Internal QA	Medium-high
8	Sources of Food Sorting Lab	74	Promising Internal QA	Medium
9	Acids, Bases & Neutralisation	73	Promising Internal QA	Medium
10	Colour Adventure	71	Promising Internal QA	Medium
11	The Breathing Process in Human	70	Promising Internal QA	Medium
12	Introduction to the Digestive System	70	Promising Internal QA	High
13	Solar System: Gravity's Orchestra	66	Needs focused improvement	High

Interpretation: 85+ = controlled-pilot candidate; 70-84 = promising Internal QA; 55-69 = focused improvement; below 55 = rebuild before pilot.

Sequence investment by trust and learning impact

Resolve release blockers first, then standardise shared quality, then invest in richer assets and outcome validation.

P0

Restore broken production experiences

Fix Solar System's roundRect runtime failure and 403 resources. Restore all 11 Digestive System GLBs. Re-audit both routes before any demonstration.

P0

Verify real rendering across devices

Reproduce the blank automated post-entry frames for Pollination, Breathing, Force and Acid/Base on Chrome, classroom PCs and Quest. Treat visibility as a release gate.

P1

Standardise narration

Create packaged, curriculum-reviewed voice assets for the eight simulations using browser speech. Keep the singleton sound manager as the global playback owner.

P1

Repair release governance

Register Colour Adventure in the canonical module contracts or remove it from the launchable list. One source of truth must determine what management can call implemented.

P1

Pilot two flagships

After Quest acceptance, run teacher-led pilots for Solubility and Pollination. Capture completion, misconception correction, teacher effort and student engagement evidence.

P2

Raise asset richness selectively

Invest first where realism teaches: anatomy, laboratory glassware/liquids, respiratory mechanics and deformation. Avoid decorative assets that do not improve understanding.

Soluble and Insoluble Substances Lab

92

OUT OF 100

PILOT CANDIDATE

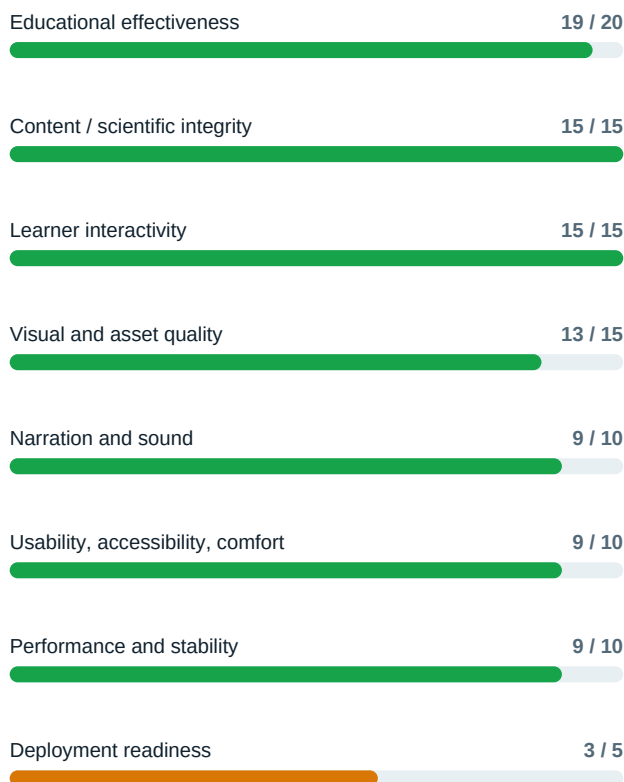
HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

The strongest current portfolio example: a visible, quantitative experiment in which students predict, measure, stir, observe, and reconcile every gram of material.

Weighted quality dimensions



WHAT IS WORKING

- Real mass balance, saturation, temperature, settling, suspension and oil-separation behaviour
- Physical beaker, 200 mL water, five materials, measured scoop and molecular evidence view
- 33 packaged narration clips plus singleton sound control prevent overlapping instructions

GAPS AND RISKS

- Most apparatus is procedural rather than authored high-detail laboratory assets
- No signed Quest headset acceptance or classroom outcome study
- Internal QA maturity means teacher workflow and multi-student robustness remain unproven

NEXT MANAGEMENT ACTION

Prioritise for the first controlled school pilot after Quest performance and teacher-facilitation acceptance.

Plant Pollination & Growth Cycle

82

OUT OF 100

PROMISING INTERNAL QA

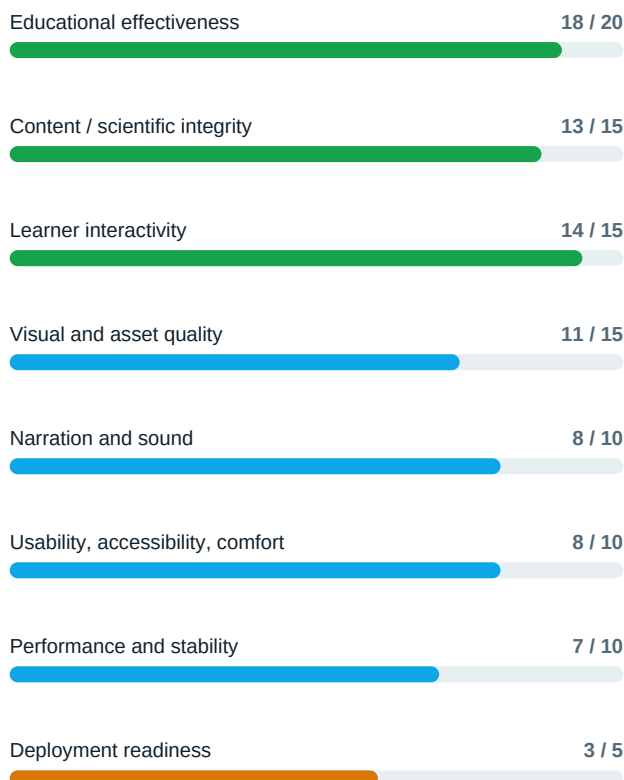
HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A well-structured investigation rather than a passive plant tour, with treatment/control reasoning and the complete journey from flower parts to germination.

Weighted quality dimensions



WHAT IS WORKING

- Eight evidence-gated stages distinguish pollination, fertilisation, seed formation and germination
- Treatment/control experiment and misconception feedback support causal reasoning
- Mapped materials, garden tools, scale disclosure, eight narration clips and 29 focused tests

GAPS AND RISKS

- Live automated entry cleared to a flat green frame, so cross-device visual reliability needs acceptance testing
- Asset richness is primarily procedural and does not yet match premium botanical simulations
- No classroom learning-gain or teacher-usage evidence

NEXT MANAGEMENT ACTION

Run a Quest/browser rendering acceptance pass, then pilot as the second flagship science experience.

States of Matter Particle Lab

81

OUT OF 100

PROMISING INTERNAL QA

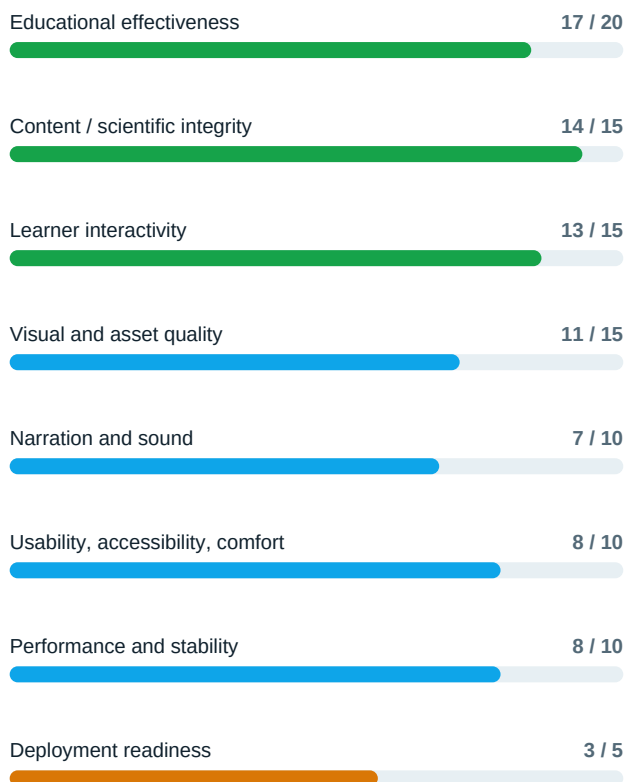
MEDIUM-HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A compact, conceptually strong particle laboratory that makes invisible spacing and motion observable while linking heat changes to phase behaviour.

Weighted quality dimensions



WHAT IS WORKING

- Authored solid, liquid and gas reference states with tested heat mapping
- Instanced particles efficiently show spacing, cohesion and motion
- Prediction/observation flow, representation disclosure and four packaged stage narrations

GAPS AND RISKS

- Only four stages limit repeated investigation and transfer
- Real-time audit remained at the mode-selection presentation after automated entry
- Particle representation is instructional, not molecularly literal, and needs explicit teacher framing

NEXT MANAGEMENT ACTION

Add a comparison/transfer challenge and complete browser plus Quest entry-flow acceptance before pilot.

Electric Circuits & Resistance (Ohm's Law)

79

OUT OF 100

PROMISING INTERNAL QA

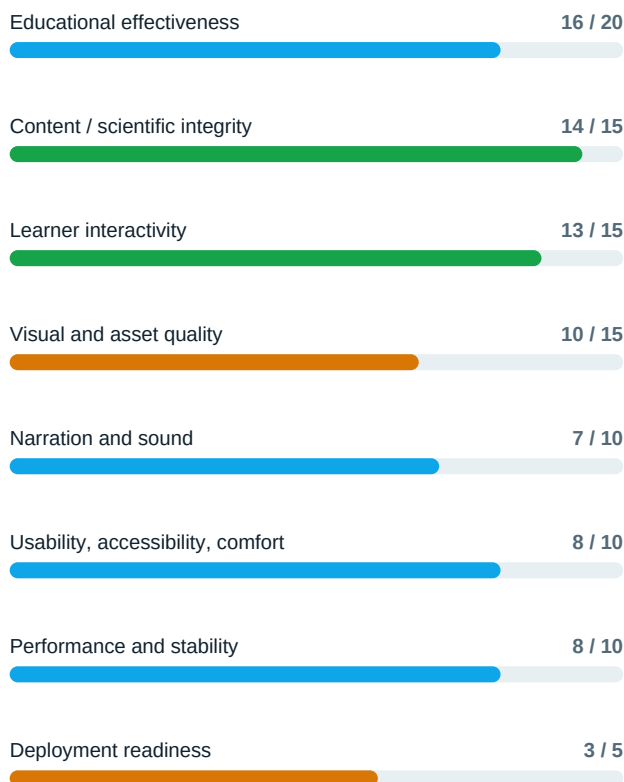
MEDIUM-HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A defensible interactive circuit model that connects physical components, current flow, resistance changes and prediction through Ohm's law.

Weighted quality dimensions



WHAT IS WORKING

- Verified electrical truth model for open/closed circuits and $V=IR$
- Learners manipulate physical battery, switch, bulb and resistor objects
- Mapped PBR presentation, four packaged stage narrations and browser/VR controls

GAPS AND RISKS

- Only four stages and limited component variety constrain experimentation depth
- Automated live entry remained on the browser/VR choice presentation
- No measurement uncertainty, multimeter workflow or classroom validation

NEXT MANAGEMENT ACTION

Extend with meter reading and fault-finding tasks, then run end-to-end controller acceptance.

Introduction to Money

76

OUT OF 100

PROMISING INTERNAL QA

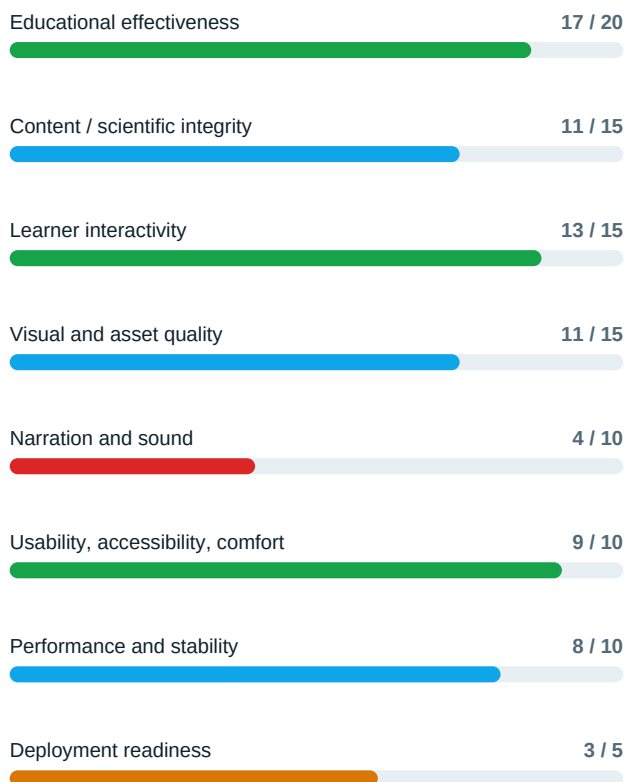
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A colourful, age-appropriate Money Town journey that moves from identifying Indian currency to comparing values and making simple purchases.

Weighted quality dimensions



WHAT IS WORKING

- Five-stage progression covers coins, notes, value comparison, shopping and memory
- Large targets, simple prompts, comfort mode and browser/VR navigation suit young learners
- Live child-facing scene rendered cleanly and dedicated lesson tests protect scoring logic

GAPS AND RISKS

- Narration depends on browser speech rather than consistent packaged voice assets
- Procedural assets look playful but not premium or culturally detailed
- Currency authenticity and classroom usability have not been externally reviewed

NEXT MANAGEMENT ACTION

Record a child-friendly voice pack and complete teacher review of currency wording and purchase tasks.

Preposition Adventure

76

OUT OF 100

PROMISING INTERNAL QA

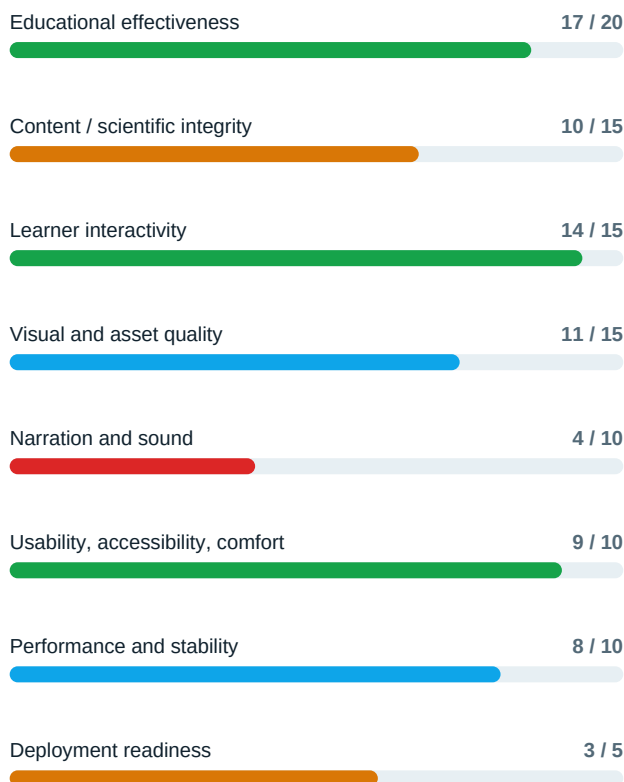
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A strong use of spatial XR for language learning: students place objects, connect position to meaning, and practise simple location sentences.

Weighted quality dimensions



WHAT IS WORKING

- Placement interaction makes abstract position words concrete
- Five stages include guided practice, sentence prompts and memory checks
- Bright scene, large controls, comfort mode and clean live rendering

GAPS AND RISKS

- Browser text-to-speech cannot guarantee consistent pronunciation or classroom voice quality
- Automated scoring emphasises placement more than spoken-language evidence
- Procedural environments repeat the visual language of other early-years lessons

NEXT MANAGEMENT ACTION

Add packaged narration and teacher-reviewed spoken-response prompts before a language-learning pilot.

Effects of Force on Motion and Shape

74

OUT OF 100

PROMISING INTERNAL QA

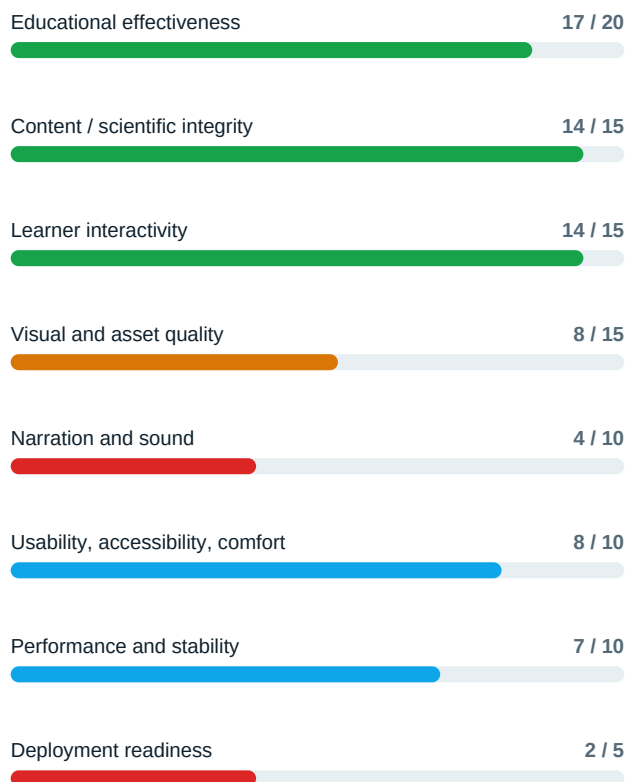
MEDIUM-HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

The physics foundation is stronger than the presentation: learners push, redirect, brake and deform objects through a deterministic motion model.

Weighted quality dimensions



WHAT IS WORKING

- Fixed-step tests demonstrate inertia, braking, directional force and bounded collisions
- Six stages connect starting, stopping, speeding, direction change and deformation
- Direct manipulation provides meaningful cause-and-effect feedback

GAPS AND RISKS

- Post-entry production audit produced a blank headless frame despite a mounted canvas
- No packaged narration and limited visual asset richness
- Deformation examples need closer visual realism and headset acceptance

NEXT MANAGEMENT ACTION

Treat visual rendering acceptance and richer deformation feedback as release gates before pilot.

Sources of Food Sorting Lab

74

OUT OF 100

PROMISING INTERNAL QA

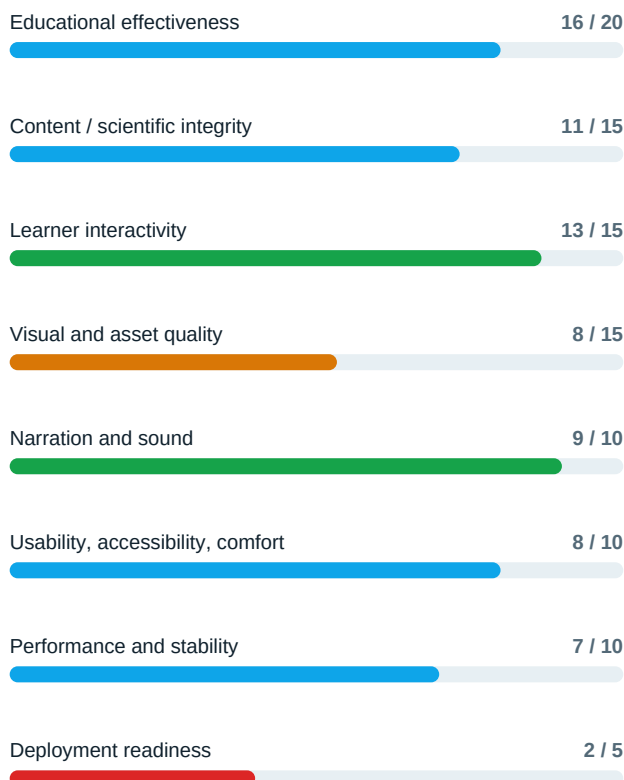
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A clear, usable sorting experience with unusually complete spoken feedback, but limited visual depth and comparatively thin dedicated test coverage.

Weighted quality dimensions



WHAT IS WORKING

- Students classify eight foods as plant, animal or fungal and receive misconception feedback
- 36 packaged clips cover stages, item clues, correct and incorrect placements
- Live tabletop and controller-accessible category platforms rendered without errors

GAPS AND RISKS

- Simplified food tokens and tabletop make the experience feel closer to a 3D worksheet
- Only two dedicated viewer regression tests despite many feedback branches
- Little transfer beyond the authored item set

NEXT MANAGEMENT ACTION

Improve food asset recognisability, add transfer items, and expand scoring regression coverage.

Acids, Bases & Neutralisation

73

OUT OF 100

PROMISING INTERNAL QA

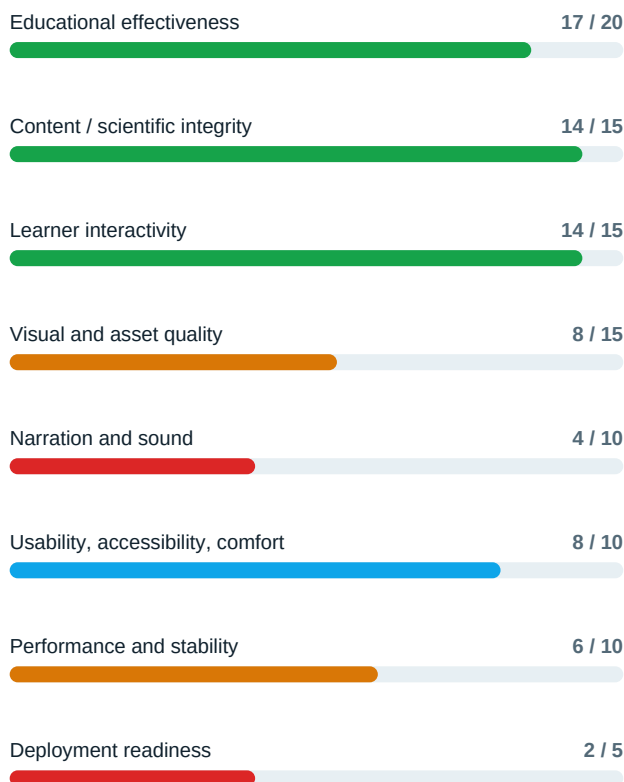
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A sound learning sequence for indicators and neutralisation, with a tested chemistry model but a presentation that still reads as an early procedural prototype.

Weighted quality dimensions



WHAT IS WORKING

- Five stages connect litmus, universal indicator, pH and neutralisation
- Tested model maps acid/neutral/base states and indicator colours
- Learners act on an experiment bench rather than only answer questions

GAPS AND RISKS

- Post-entry production audit produced a blank headless frame despite a mounted canvas
- No packaged narration or high-detail glassware/liquid assets
- No explicit laboratory-safety assessment or classroom validation

NEXT MANAGEMENT ACTION

Resolve cross-browser rendering, add laboratory-quality materials and include a safety decision stage.

Colour Adventure

71

OUT OF 100

PROMISING INTERNAL QA

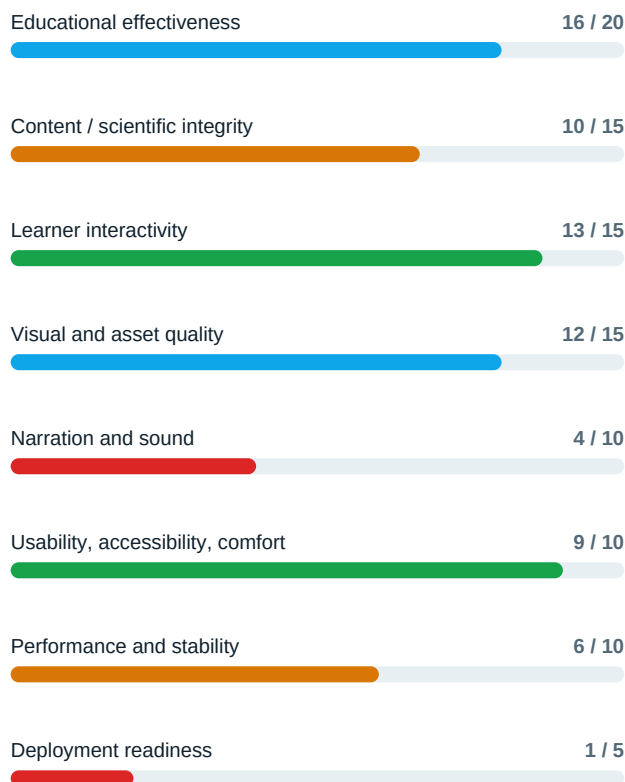
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A cheerful early-years colour experience with good visual appeal and gated memory activities, weakened by release-governance and narration gaps.

Weighted quality dimensions



WHAT IS WORKING

- Bright rainbow classroom and colour worlds are immediately understandable
- Five-stage recognition, finding and memory journey uses large age-appropriate controls
- Live scene rendered cleanly with comfort and restart support

GAPS AND RISKS

- Launchable publicly but absent from the canonical 12-module release registry
- Relies on browser speech instead of a consistent child-friendly voice
- Scientific/content review evidence is lighter than for curriculum-registered modules

NEXT MANAGEMENT ACTION

Fix release registration first, then package narration and conduct an early-years usability review.

The Breathing Process in Human

70

OUT OF 100

PROMISING INTERNAL QA

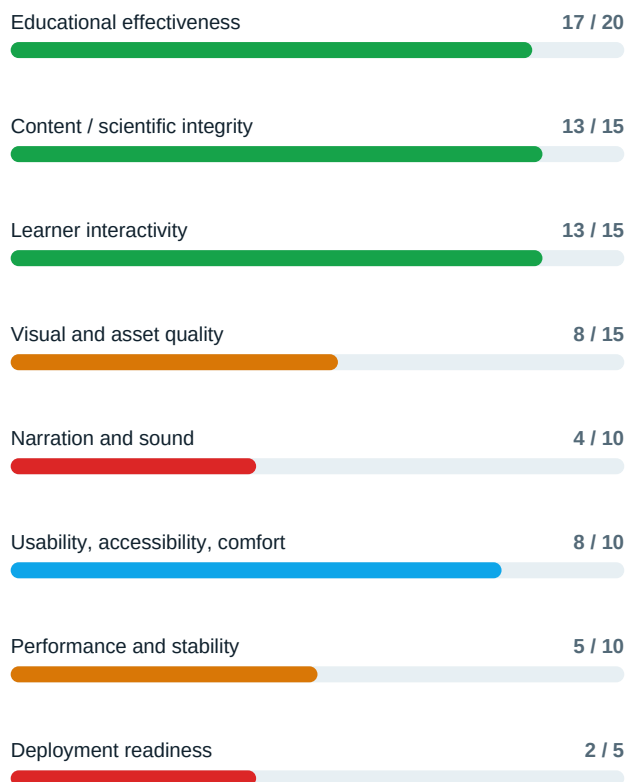
MEDIUM INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

A well-scoped six-stage physiology lesson covering both mechanics and gas exchange, but current production rendering and asset evidence are not yet convincing enough for pilot use.

Weighted quality dimensions



WHAT IS WORKING

- Traces airway, lungs, diaphragm, chest volume and alveolar gas exchange
- Action/evidence gates prevent passive progression
- Low comfort-risk design and shared browser/VR framework

GAPS AND RISKS

- Post-entry production audit produced a blank headless frame despite a mounted canvas
- Procedural anatomy and no packaged narration reduce realism and consistency
- Only four dedicated experience tests and no external anatomy review evidence

NEXT MANAGEMENT ACTION

Prioritise visible respiratory mechanics, packaged narration and anatomy review before headset trials.

Introduction to the Digestive System

70

OUT OF 100

PROMISING INTERNAL QA

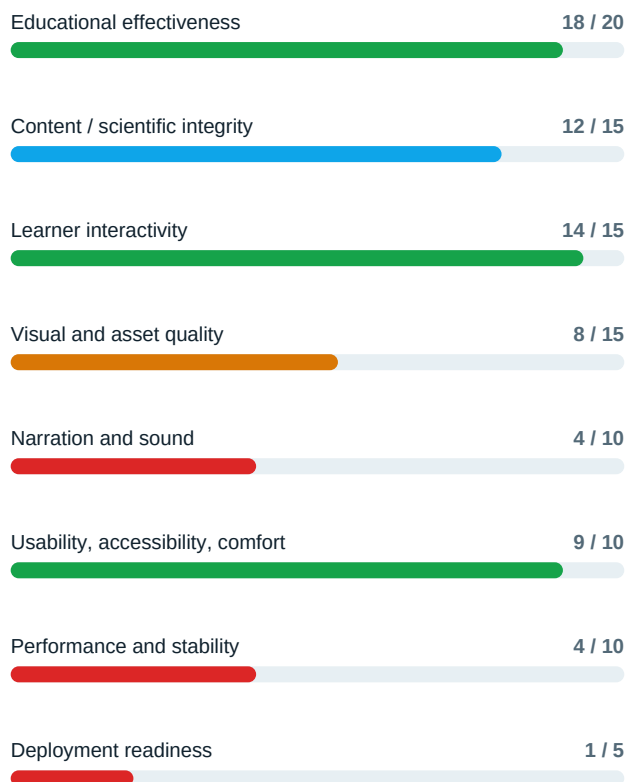
HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

The lesson design is ambitious and thorough, but the deployed visual experience is materially degraded because every authored organ model request fails and falls back to procedural geometry.

Weighted quality dimensions



WHAT IS WORKING

- Ten-stage food journey covers organs, peristalsis, accessory organs, absorption, recap and healthy habits
- Strong accessible HTML controls, subtitles, comfort mode and Quest controller design
- 21 focused tests protect pathway, evidence gates and assessment scoring

GAPS AND RISKS

- All 11 production GLB requests returned 404 during audit
- No packaged narration; browser voice quality varies by device
- Procedural fallback cannot deliver the intended body-scale anatomical realism

NEXT MANAGEMENT ACTION

Block pilot release until GLB delivery is fixed and every organ is visually accepted on browser and Quest.

Solar System: Gravity's Orchestra

66

OUT OF 100

NEEDS FOCUSED IMPROVEMENT

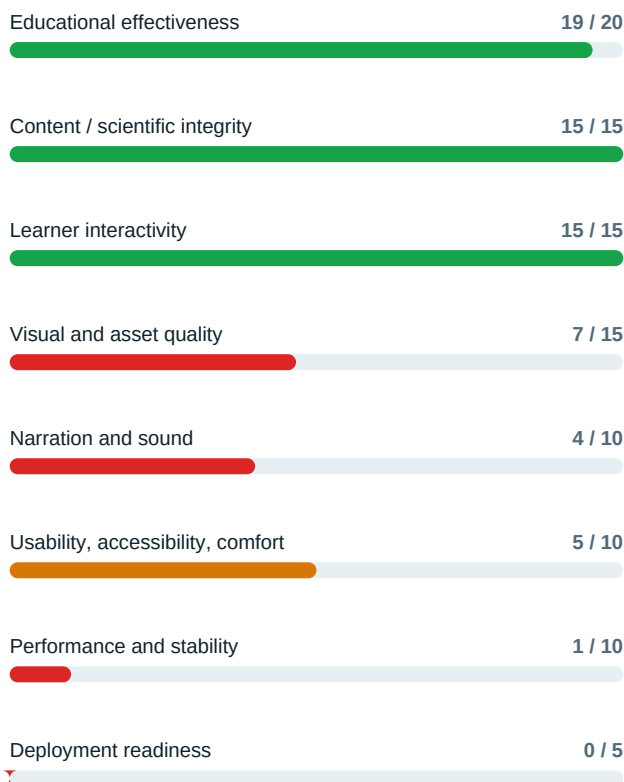
HIGH INTERNAL EVIDENCE

Release maturity: INTERNAL QA

MANAGEMENT SUMMARY

The portfolio's deepest conceptual and scientific design is currently unusable in production: the live route pauses before mounting the 3D world.

Weighted quality dimensions



WHAT IS WORKING

- Eight-stage mission confronts orbital-speed, hottest-planet, scale and comet misconceptions
- Verified Kepler-law, orbital-period, scale-blend and comet models with 35 dedicated tests
- Credited local textures for the Sun and all eight planets plus a free observatory mode

GAPS AND RISKS

- Production stops with 'roundRect is not a function' and mounts no canvas
- Two production resources returned 403 in the focused runtime inspection
- No packaged narration and medium comfort risk remain unvalidated on Quest

NEXT MANAGEMENT ACTION

Treat as a release blocker: fix the runtime and resource failures, redeploy, then repeat the full visual/audio/controller audit.

How to interpret these quality scores

This audit is deliberately conservative: absent evidence is treated as unverified, and live failures override implementation ambition.

Weighted rubric

Educational effectiveness	20 points
Content / scientific integrity	15 points
Learner interactivity	15 points
Visual and asset quality	15 points
Narration and sound	10 points
Usability, accessibility, comfort	10 points
Performance and stability	10 points
Deployment readiness	5 points

Evidence used

- Production route, canvas, console and screenshot audit
- Launch registry and canonical simulation modules
- Viewer, lesson, world and scientific-model source
- Packaged audio and public asset inventory
- Dedicated regression, pedagogy and model tests
- Current release maturity and deployment status

LIMITATIONS

- No controlled classroom learning outcomes, teacher workload data or signed Quest acceptance runs were available.
- Automated headless rendering can expose cross-browser problems but does not replace physical-headset inspection.
- Narration technical quality was measurable; voice naturalness and child comprehension were not formally listener-tested.
- Scores compare the current portfolio and should be recalculated after fixes, asset delivery and field evidence.

AUDIT-SPECIFIC FINDINGS

- The public launch list contains 13 simulations, while canonical release contracts recognise 12; Colour Adventure is the unmatched entry.
- The solar production route stops with 'roundRect is not a function' before mounting the 3D canvas.
- The digestive production route returns 404 for all 11 authored GLB requests and uses procedural fallback geometry.
- Packaged narration exists for Pollination, Circuit, States of Matter, Food Sources and Solubility; the remaining eight use device speech fallback.